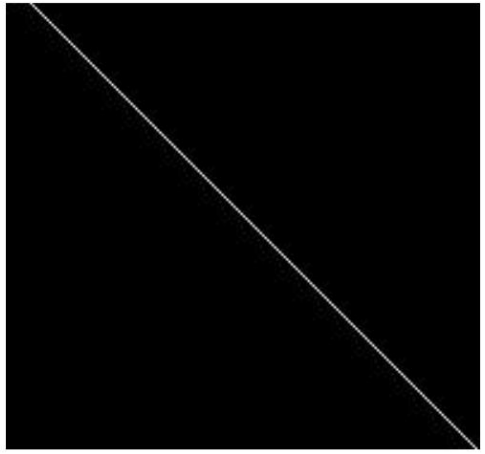
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City University of Hong Kong
(Dongguan)



Image with Edge



Edge Location

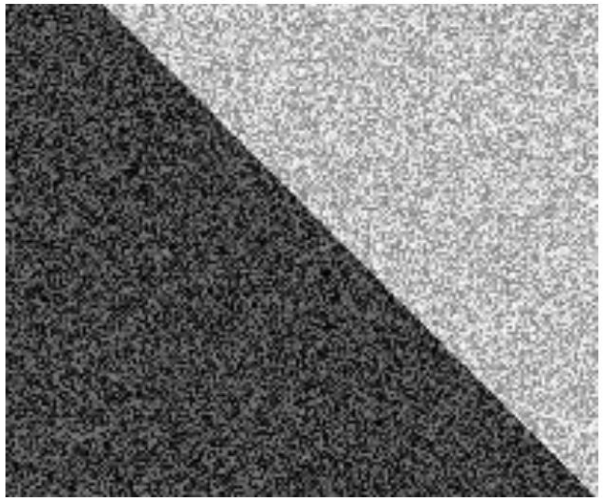
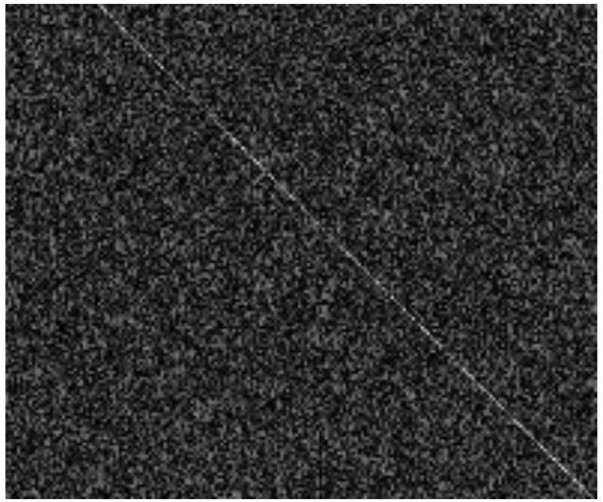
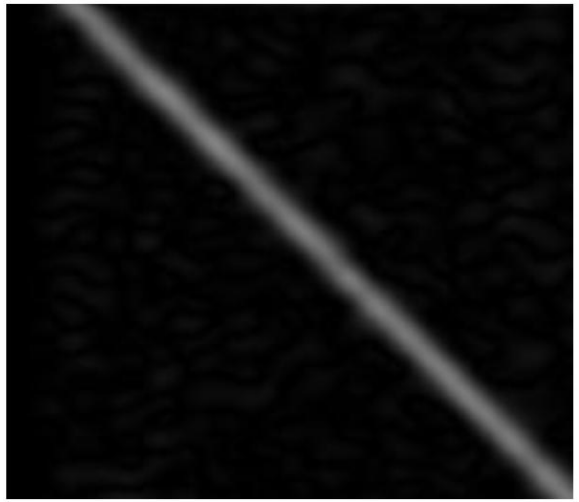


Image + Noise



Derivatives detect edge *and* noise



Smoothed derivative removes noise, but blurs edge

Example



original image

Demo: http://bigwww.epfl.ch/demo/ip/demos/edgeDetector/

Image credit: Joseph Redmon

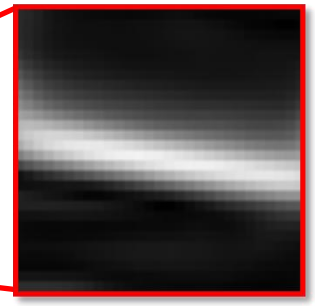
Finding edges



smoothed gradient magnitude

Finding edges





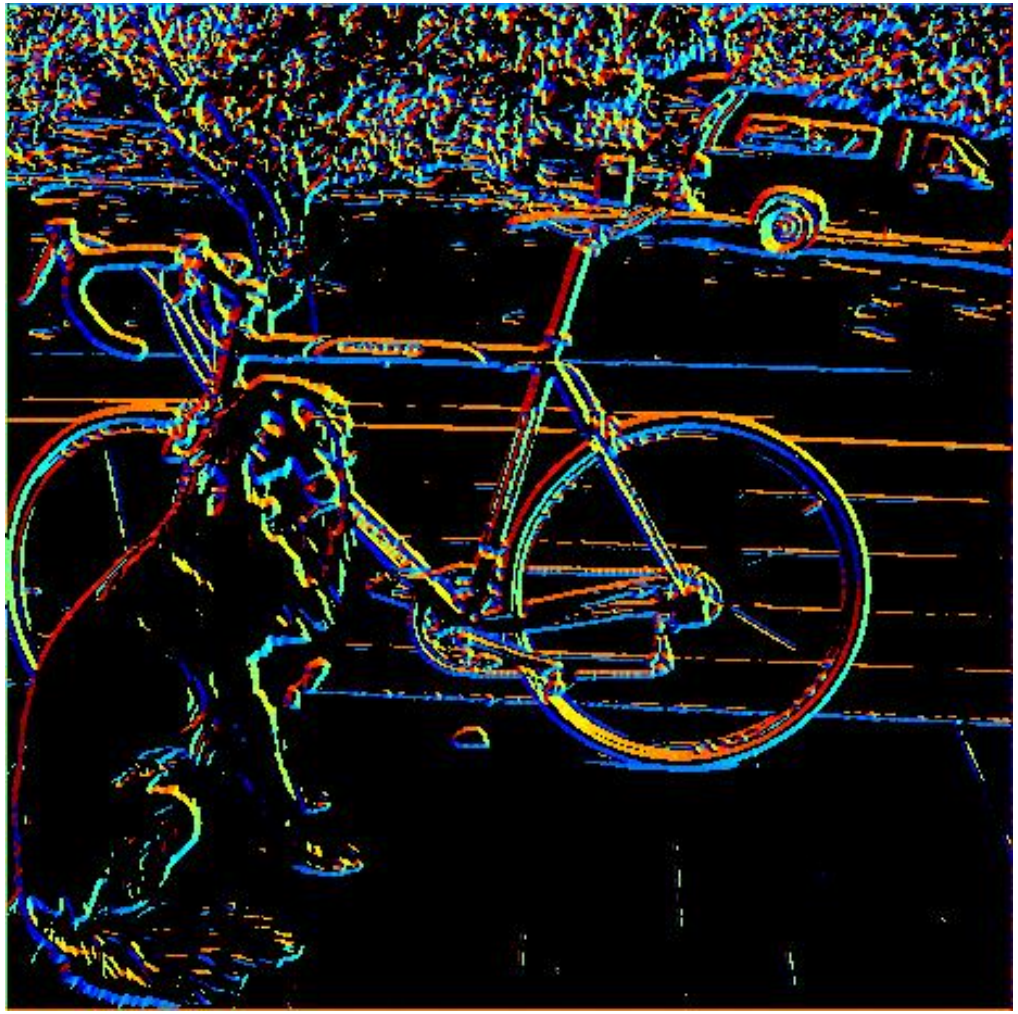
where is the edge?

thresholding

Get Orientation at Each Pixel

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- Get orientation (below, threshold at minimum gradient magnitude)

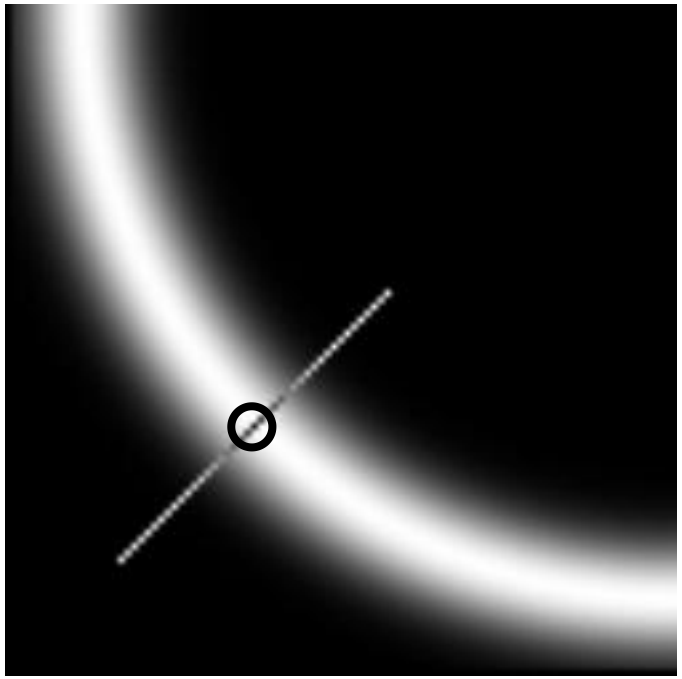


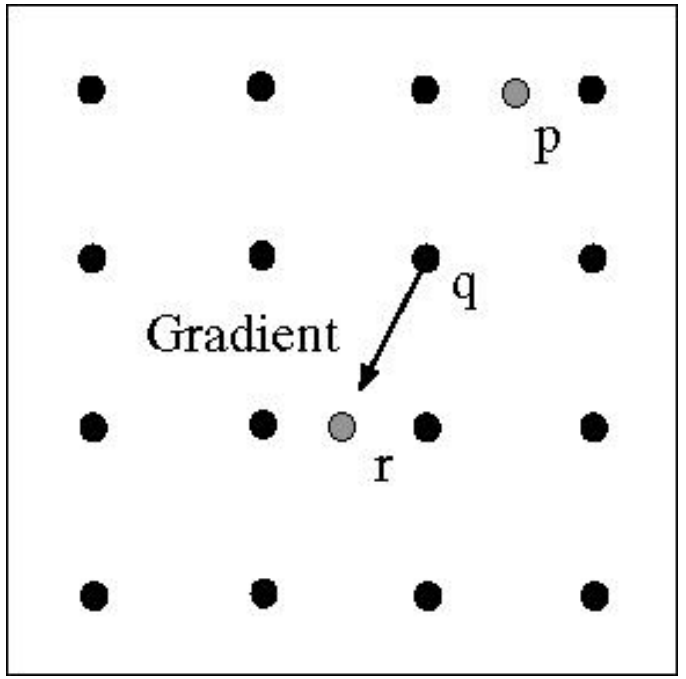
$$\theta = \text{atan2}(\text{gy}, \text{gx})$$

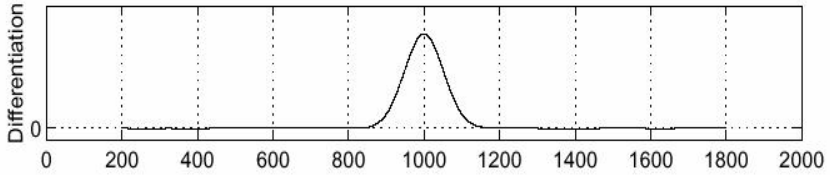


Gradient orientation angle

Non-maximum supression







- Check if pixel is local maximum along gradient direction
  - requires *interpolating* pixels p and r

Before Non-max Suppression



After Non-max Suppression

